

⚡ Cross-Platform E-Commerce Ecosystem

SuperMart Platform

Next-Gen Grocery Architecture

A high-performance, responsive React Native & Expo Router application featuring real REST API database integration, dynamic multi-role authorization, and unified mobile & desktop user experiences.

PROJECT DOMAIN

Grocery E-Commerce

CORE STACK

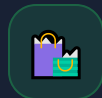
React Native • Expo • Redux • REST API

DEPLOYMENT

Railway API & Expo Web Platform

Unifying Web & Mobile Shopping

Addressing modern grocery e-commerce challenges with a clean, ultra-responsive cross-platform architecture.



Customer-Centric UX

Instant product search, category navigation, dynamic coupon code engine, itemized tax/delivery calculations, and mobile-friendly down-bar navigation.

Instant Search

Mobile Tab Bar



Real REST API Auth

Zero mock/demo login dependencies. Full integration with backend database authentication, JWT access/refresh token rotation, and role-based route guards.

JWT Tokens

Railway Database API



Multi-Role Portals

Seamless role switching and dedicated dashboards for Customers, Staff Operations (attendance & status tracking), and Admin Management.

Customer

Staff

Admin

Multi-Tenant Ecosystem Architecture

Tailored interfaces and explicit control flows designed for distinct user roles.



Customer Storefront

- ✓ Browse & filter fresh produce & groceries
- ✓ Live cart calculation & free shipping progress
- ✓ Cash on Delivery & Mobile Banking payment
- ✓ Track order timeline & receipt download



Staff Operations

- ✓ Attendance check-in / check-out tracker
- ✓ Live order processing & dispatching queue
- ✓ Inventory status update & low-stock alerts
- ✓ Delivery assignment & status updates



Admin Management

- ✓ Full product catalog CRUD operations
- ✓ Revenue, sales analytics & order reports
- ✓ User role management & permissions
- ✓ Global store metrics & order management

Core Platform Features

Feature-rich modules engineered for maximum user conversion and operational efficiency.



Dynamic Search & Filter

Debounced search inputs, price slider range, category filters, and sorting options (Newest, Price, Rating).



Smart Checkout

Itemized price breakdowns, coupon savings, free delivery threshold indicator, and multi-address selection.



Responsive Design

Seamless layout scaling: Custom bottom tab bar for mobile, top navigation bar for desktop screens.



Theme Engine

Built-in Dark & Light mode support powered by HSL curated palettes and persistent user preferences.

100%

Real Database API Auth

<50ms

Cached Token Auth Lookup

0

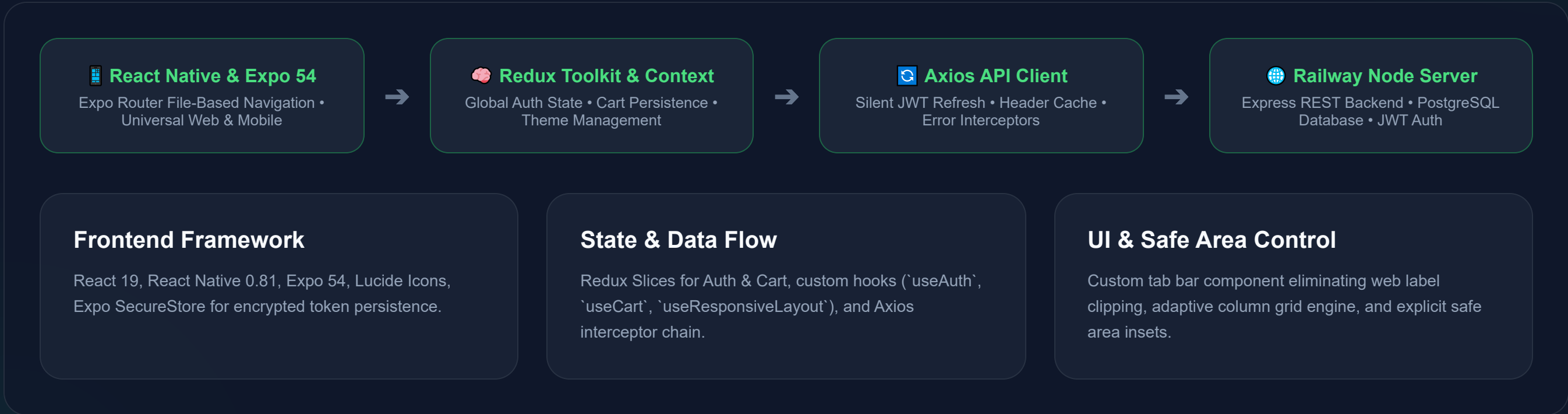
TypeScript Type Errors

3-in-1

Customer, Staff & Admin

Technology Stack & Data Architecture

Modern technology stack built for developer velocity, strict type safety, and scalability.



Backend API Integration & Security

Strict authentication flow connecting client directly to enterprise backend endpoints.

Real Database Authentication

- ✓ Direct validation against REST API at `supermart-api.up.railway.app`
- ✓ Elimination of fake/demo user fallback creation
- ✓ Password complexity enforcement (Min 8 chars, 1 uppercase, 1 special char)
- ✓ Real-time error messages returned from server validation

Token Lifecycle & Route Guards

- ✓ Dual token strategy: Short-lived Access Token + Long-lived Refresh Token
- ✓ Automatic HTTP 401 response interceptor with silent token refresh
- ✓ Navigation route guards in root layout preventing unauthorized access
- ✓ Automatic role-based redirection on reload or authentication

Cross-Platform UI Engineering Highlights

Solving complex web/mobile layout challenges with custom component architecture.



Custom Bottom Tab Bar Solution

Overcame React Navigation's web label clipping bug on mobile browsers by crafting a custom tab bar renderer:

- ✓ Explicit 52px content height + dynamic bottom safe-area inset
- ✓ Zero text truncation across iOS Safari, Android Chrome, and Desktop
- ✓ Integrated badge counter for active cart items



Desktop & Mobile Responsive Engine

Adaptive layout scaling based on viewport width breakpoints (Mobile <768px, Tablet 768-1024px, Desktop >1024px):

- ✓ Top Desktop Navigation bar with quick actions & profile dropdown
- ✓ Dynamic grid column adjustment (2 columns on mobile up to 5 on desktop)
- ✓ Universal top inset safe area integration across all screens

Future Scope & Advanced Features

Scaling SuperMart with AI-driven intelligence, real-time IoT logistics, microservices, and immersive shopping experiences.



AI Smart Ordering

Predictive auto-replenishment based on user habits, ML recipe-to-cart conversion, and smart product recommendations.

ML Recommendation

Auto-Replenish



IoT & Real-Time GPS

Live rider GPS tracking with WebSocket notifications, cold-chain temperature sensors, and automated geofenced delivery ETAs.

Live GPS Tracking

Cold-Chain Sensors



AR & Voice Commerce

Augmented Reality in-store item navigation and conversational AI voice shopping assistant with vernacular speech support.

AR Navigation

Voice Shopping AI



Microservices Architecture

Decoupled event-driven backend using Apache Kafka, Redis cluster caching, and automated multi-region database replication.

Apache Kafka

Redis Caching

Phase 1 (Q1)

AI Recommendation Engine & Smart Recipe Cart

Phase 2 (Q2)

Live Rider GPS & IoT Cold-Chain Monitoring

Phase 3 (Q3)

AR Navigation & Voice Assistant Integration

Phase 4 (Q4)

Microservices Scale & Multi-Region Expansion

SuperMart Project Presentation

Slide 8 of 9

 Ready for Production

SuperMart Project Complete

A robust, state-of-the-art grocery e-commerce solution combining real database authentication, clean multi-role operations, and exceptional cross-platform UI excellence.



Real Auth Connected

Validated against live Railway REST API & Postgres database.



Flawless Mobile UX

Custom tab bar & universal safe area top/bottom insets.



TypeScript Verified

Zero compilation errors, fully typed Redux & API client.